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Comprehensive private schooling for low-income children: Experimental case-study evidence from Mexico☆

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ABSTRACT

We use first-grade lottery-based admissions to estimate impacts and cost-effectiveness of a subsidized comprehensive private school for low-income children in Mexico City, part of a philanthropic organization supporting and operating similar schools worldwide. Relative to students who did not win the lottery, CHM lottery winners gain additional 0.18 SD in literacy and 0.09 SD in numeracy over the first three years of elementary school. Parents of lottery winners are more likely to report children's school is academically demanding, rate the school higher and have greater expectations of children's college completion. Achievement gains come at an increased cost relative to counterfactual public schools of $1000/pupil-year, which suggests low cost-effectiveness. Higher cost is explained by greater array of services and few economies of scale. Despite the high per student cost, this robust case study suggests philanthropic private schools have great potential to improve achievement amongst the region's most vulnerable students and reduce longstanding learning and opportunity gaps.

1. Introduction

For close to three decades Mexico has had near universal primary enrollment, yet most children who complete primary school are not academically prepared for high school (Kattan and Székely, 2014). Results from the 2018 PISA test show that over a third of 15-year old students in Mexico did not achieve a minimum level of competency in reading, math, and science (OECD 2019b). In the past two decades, the achievement trend in PISA for Mexico has been mostly flat and disparities persist: socio-economically advantaged students outperformed disadvantaged students in reading by 81 score points—a difference of up to two years of schooling (OECD 2019, 2019b). Poor academic preparation is one of the key reasons only about half of Mexican youth complete secondary school –compared to 85% among developed nations (OECD, 2019; Kattan and Székely, 2014).

As in many other countries, those Mexican parents who can afford it turn to the private (non-government) sector to seek higher quality schools (Heyneman and Stern, 2014). Latin America is one of the regions with greatest private sector participation in the world with 20% of elementary and 21% of secondary school students enrolled in private schools (Elacqua et al., 2018). While Mexico is an exception, private school enrollment in the country is on a rising trend. According to UNESCO data, seven percent of primary school students in Mexico were enrolled in private schools in 2000; that number is now close to 10%.[1] Private schools are an attractive option because, compared to public schools, they offer a longer school day and a larger, more diverse slate of extracurricular and enrichment offerings, including English as a second language. Most Mexican parents, however, cannot afford to pay for private school tuition which ranges from US$160-$600 per month, on average. According to national statistics close to half of the country's

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[1] Data available at: https://data.worldbank.org/indicator/SE.PRM.PRIV.ZS?locations=MX

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population earns an income below the poverty line (around $300 US per month in urban areas).[2] There are no charter schools in Mexico. Therefore, families at the lower end of the income spectrum have little school choice. For low income families in Mexico, as is the case in many other countries in Latin America and around the world, philanthropically financed private schools provide access greater school quality at little to no tuition cost (Heyneman and Stern, 2014).

Most of the experimental literature on private schools focuses on lower-income countries such as Pakistan, India, Kenya, and Uganda. In these countries, studies generally find that private school options increase academic achievement and other outcomes such as gender gaps at a fraction of the public sector cost (Patrinos et al., 2009; Muralidharan and Sundararaman, 2015; Andrabi et al., 2006; Alcott and Rose, 2016; Heyneman and Stern, 2014; Bizenjo, 2020).

Two philanthropic school models that have received some attention from scholars are "*Fe y Alegria*" and "Concession" schools. "*Fe y Alegria*" schools educate low-income children in 15 countries in Latin America and "Concession" schools educate thousands of children in Colombia. These privately managed, philanthropically-funded schools offer comprehensive resource-rich schooling to low-income children at either no cost, or a fraction of the cost. Non-experimental studies of these two programs suggest students have better educational outcomes relative to students in public schools (Barrera-Osorio, 2005; Allcott and Ortega, 2009). While these models have been around for decades, we know little about the impact of non-profit, philanthropic, low-tuition fee schools on academic and other outcomes.

This paper aims to fill this gap by examining impact and cost-effectiveness of a comprehensive, philanthropic private school model targeting low-income urban children in Mexico City. We present results from a randomized control trial case study of *Christel House de México* (CHM henceforth) in Mexico City that takes advantage of first-grade lottery-based admissions to assess the school's impact on student learning, parental perceptions and expectations, and costs relative to attending local public schools.

To measure learning outcomes in 1st-3rd grade, we use the Early Grade Reading Assessment (EGRA) and Early Grade Mathematics Assessment (EGMA), which we psychometrically validate and apply to the study sample over three years.[3] Over three years, first-grade admissions lottery winners score 0.18 standard deviations (SD) higher in literacy and 0.09 SD higher in numeracy than students who did not win the CHM lottery. Early-grade literacy outcomes have been found to be predictive of better school performance in high school (Cunningham and Stanovich, 1997).

We also find evidence that parents of lottery winners are more likely to report their children's school is academically demanding, rate the school higher, and have greater expectations of children's college completion. High parent expectations highlight a potential avenue for intergenerational social mobility since fewer than two percent of parents who apply to CHM attended college. Research indicates that high parental expectations can positively influence academic outcomes for low-income students (Domina, 2005; Marschall, 2006; Yamamoto and Holloway, 2010).

Private schools serving low-income children in developing countries serve an important function in improving access and quality for all students, yet they are frequently financially at risk because they lack a secure source of funds (Heyneman and Stern, 2014; Singh and Bangay, 2014). To understand the cost of providing CHM services and the cost-effectiveness of our estimates we do a cost calculation using best-practice ingredients methods (Dhaliwal et al., 2011; Levin and McEwan, 2002, 2000). We account for direct costs to provide itemized CHM's services, as well as CHM parents' out-of-pocket expenses compared to parents in public schools. The implied cost effectiveness ratio at the margin indicates that annual achievement gains cost an additional $1000 per pupil-year relative to the costs of counterfactual public schools. This contrasts with other developing country evidence, which suggests private schools are usually cheaper to operate than public schools (Alcott and Rose, 2016; Andrabi et al., 2006; Bizenjo, 2020; Heyneman and Stern, 2014; Muralidharan and Sundararaman, 2015; Patrinos et al., 2009; Singh and Bangay, 2014 (Barrera-Osorio et al., 2017)).

Even though our study presents results from a single case study, it nonetheless provides valuable evidence for two main reasons. First, this is the first study to our knowledge, to experimentally test the impact on student learning outcomes of attending a philanthropically funded, comprehensive private school that specifically targets low-income children. Given the longstanding school achievement gaps evident in Mexico and the difficulties inherent in educating children living in poverty, lessons from this study could inform similar efforts in the region. Second, we generate cost-effectiveness results by carefully documenting costs through best-practice ingredients method (Levin and McEwan, 2000), a calculation that is often lacking in these types of studies.

2. The Mexican education system and CHM: a brief overview

Close to 90% of elementary and lower secondary students in Mexico attend public schools (OECD, 2014). In the most recent standardized test administered to all 6th graders in the country close to 50% of students did not reach proficiency in Spanish and 61% did not reach proficiency in Mathematics (INEE, 2015).

As is the case around the developing world (Ganimian and Murnane, 2016), most public schools in Mexico meet for half a day (4.5 h) due to infrastructure capacity constraints. They follow centrally-set national curriculum standards. Most offer some kind of preventive health services, remedial and extracurricular activities. None of these services are required by law. Public school parents in Mexico routinely perform volunteer cleaning and maintenance services (Santibañez, O'Donoghue & Abreu, 2014). Teachers are hired centrally by the federal (or state) government and are granted tenure after 6 months on the job. Over 90% of national expenditures on education go toward paying staff payroll (OECD, 2013). Public schools in Mexico are organized into elementary (grades 1–6), middle (grades 7–9) or high schools (grades 10–12). In Mexico and most of Latin America, privatization efforts are highly controversial and charter schools are not allowed largely due to formidable opposition from strong teachers' unions (Hecock, 2014; Santibañez and Jarillo, 2008).

To enroll in a public school, parents apply to Mexico City's office of education and list their top three choices—which can include traditional half-day schools or full-day schools. Parents are given preference when their residential or work place is in the school's catchment zone. Charter schools are not legally allowed, and neither are vouchers at any educational level.

2.1. Christel Organization and Christel House de México (CHM)

CHM is part of the Christel House International Organization (CHI), a philanthropic educational initiative that provides educational services to over 5000 socioeconomically disadvantaged children around the world, including India, South Africa and the United States, and Jamaica (CHI, 2019). During the time of our analysis CHM served grades 1–9, but has since expanded to serve grades 1–12. CHM is a "community school" providing free or heavily subsidized private education and wraparound services to over 400 low-income children. During 2013, the last year that Mexico held a national standardized exam testing all students in all grades, CHM 3rd graders scored higher than the Mexico City average:

[2] https://www.coneval.org.mx/Medicion/Paginas/PobrezaInicio.aspx

[3] EGRA and EGMA are a battery of literacy and numeracy tests developed for the United States Agency for International Development and used in over 50 countries. Mexico does not have a standardized testing programs for grades below 3rd grade.

661 points in Mathematics (vs. 619), and 598 points in Spanish (vs. 586).

CHM is a resource-rich institution offering services above and beyond what public schools offer (see Table 1). These include a longer school day, English as a second language, a tutoring program, counseling for career- and college-readiness, etc. Consistent with CHM's comprehensive community model, students receive "wraparound" services including preventative health care—mental health services, annual medical, and dental check-ups—and a nutritional package that includes daily breakfast and lunch following healthy dietary guidelines. All children receive school uniforms and school materials free of charge.

CHM follows the national curriculum standards set by the Mexican Ministry of Education. Teachers are not unionized and do not have tenure. CHM teachers have less seniority and earn lower salaries than teachers in public schools and teachers can be hired/fired at will. CHM is able pay teachers less than comparable public schools by drawing from the pool of teachers who graduate from private teacher colleges or traditional universities or retired public-school teachers who receive a pension and are thus able to supplement lower earnings.

Table 1
Christel House de Mexico and Public Schools - Main Features.

	CHM	**Traditional Public Schools (Half-day)**	**Extended Day Public Schools "*Jornada Ampliada*"**	**Full-Time Public Schools "*Escuelas Tiempo Completo*"**
Length of school day (hours)	9	4.5	6	8
Curriculum/1	National+	National	National	National
English as a second language,	Yes	Yes	Yes	Yes
Breakfast/2	Yes	Yes	Yes	Yes
Lunch	Yes	No	No	Yes
Art, music, computer skills	Yes	Some schools	Yes	Yes
Autonomy over teaching hiring/firing	Yes	No	No	No
Teacher Seniority (average, 1st-3rd grade) (average)	2	8	8	8
Parenting programs, psychological services for students/parents	Yes	No	No	No
Extended day program, tutoring, socio-emotional development, health preventive services (wraparound services)	Yes	Some schools	Yes	Yes
Students per teacher (average)	27	27	27	27

/1 CHM uses the national curriculum provided by the Secretaria de Educacion Publica (SEP), but supplements with other textbooks, instructional materials and resources
/2 Only in pubic schools that participate in the National Free Breakfast Program offer breakfast (including traditional half-time and extended day). These are usually schools that enroll a majority of low-income students.
/3 Salaries are for a 4.5 h (traditional) teaching position. In schools where teachers teach an extended day, or multiple positions, salaries will be higher.
Source: For CHM, cost interviews conducted as part of this study. For public schools, various published sources and interviews with a small sample of schools. See cost section for details.

Even though CHM offers many services to students and their families, it is not unique among other philanthropically funded schools in Mexico. A simple comparison of CHM's services relative to a convenience sample of private schools in Mexico City and other large metropolitan areas indicates that CHM offers a similar range of services as comparable philanthropic schools, except for three key areas: it formally requires significantly more parental commitment, it is the only school we found that offers lunch and breakfast to all students, and it charges lower tuition fees (see Table A1 of the Online Appendix). This highlights that, while unique and resource-intensive, variants of CHM's schooling model already exist in Mexico.

3. Research design and data

This study experimentally tests the impact of attending CHM on student test scores. To do this, we first, conduct a lottery to randomly determine who wins a spot in CHM. Second, we conduct balance and attrition tests to ensure that the "treatment" and the "control" group are similar in baseline characteristics. We then test for differential attrition from the study. There are many reasons a study can have differential attrition: families change residence or schools, students are absent when test administers come to their school, families opt-out of the study, etc. To ensure that sample attrition does not compromise the validity of the findings, we conduct several standard tests. After we can be reasonably certain that differential attrition is not a threat and our sample is balanced, we run the main analysis of the impact of CHM on student test scores using regression analysis. To do this we use lottery offers as the random explanatory variable instead of the problematic (endogenous) enrollment variable. Enrollment is problematic because it is not random, i.e. students who decide to enroll or not enroll in CHM may be different in unobserved ways. As is standard in program evaluations we estimate Intent-to-Treat effects (ITT), which means that those who receive a lottery offer are the "treatment" group—independent of whether or not they enrolled in CHM– and those who did not receive an offer are the "control" group. To translate the ITT into a more meaningful metric such as annualized achievement gains, we use instrumental variables (IV) to estimate the effect of each year spent in CHM. The research plan and data collection protocols were approved by the Institutional Review Boards of RAND Corporation and the University of Southern California.[4] Fig. A1 in the Online Appendix depicts key evaluation activities.

3.1. Lottery Procedures

Every year CHM has more eligible applicants than open seats for its 1st grade cohort. For this study, CHM agreed to use lotteries for admission to oversubscribed first-grade seats. Because its mission is to educate low-income children, CHM kept the income verification and eligibility process intact. Only income-eligible families could sign up for the lottery. CHM conducted public lotteries for the three admission cohorts we study, 2013–2015. Invited members of education NGOs observed the process. Parents chosen at random served as additional observers to verify that all procedures followed approved guidelines.

3.1.1. Student recruitment

Starting in January-February, CHM advertised through its network of social workers and community outreach contacts to encourage parents to apply to the school. In 2013, the first year of our study, CHM distributed information in kindergartens and preschools in areas of the city that concentrate low-income populations and are near CHM. These include the districts of *Alvaro Obregon*, *Magdalena Contreras*, and *Benito Juarez* (see Fig. A2 in the Online Appendix), with a total resident population of about 2 million. The timing of recruitment for CHM aligns

[4] Data collection for the study was carried out while both researchers were at these institutions.

with public school pre-registration season. Each February parents of Kindergarten students in Mexico City apply and pre-register their children in public school.

Between 2013 and 2015, 242 students in the three study cohorts participated in the lottery. CHM carried out separate lotteries for each cohort and within each cohort, separately for boys and girls, effectively creating a stratified randomization design with strata defined by the interaction of 1st grade cohort and gender. Through the lottery mechanism, 100 students, belonging to three cohorts, were offered a seat in CHM's 1st-grade cohort; 142 were not admitted. As a result of cohort-by-gender stratification, of the students offered a spot, 50% were girls and 50% were boys. In each lottery cohort, CHM also randomly selected 10 students (5 boys, 5 girls) for a wait-list. Children in the random wait list were offered a spot whenever a lottery winner did not enroll in CHM or dropped out. For the analysis, we classify wait-listed children as control (not treated). Results are similar when children in the waitlist are classified as treated (not reported for brevity, but available upon request).

3.2. Data

3.2.1. Baseline data

In order to verify income-eligibility for CHM, all families are required to submit a detailed application containing income information (monthly household income from all sources), information on their housing situation and conditions (i.e. whether home is rented, has a kitchen, etc.), number of members of the household, whether the parent is single, and other demographic and socio-economic information including parental occupation and education level. The application process requires all prospective students to be tested using the RAVEN progressive matrices test which measures abstract reasoning and can be used as a non-verbal estimate of baseline cognitive ability. In this study we found that RAVEN also has predictive power over EGRA/EGMA outcomes, helping improve statistical power.

3.2.2. Lottery data

We collect lottery outcome data and subsequent school enrollment. Records were frequently updated to account for sample attrition. Overall, treatment compliance was high. Of those originally offered a 1st grade spot, 93% took up the offer and enrolled in CHM. Among those not offered a spot (control students), 8.5% subsequently enrolled in CHM – most of these students came from the waitlist. After the lottery was conducted, students enrolled in either CHM or a public school. Of the total sample, 39% enrolled in CHM, 26% enrolled in full-day public schools, and 33% enrolled in half-day public schools. An additional two percent enrolled in schools of unknown type.

We collected socio-economic data at baseline (before the lottery) and achievement and parental perception data at the end of each academic year. Each cohort was followed from lottery until the last data collection round in 2016. As a result, there are three years of data for the 2013 first cohort (i.e. data for 1st, 2nd and 3rd grade), two years of data for the 2014 cohort (1st and 2nd grade), and one year of data for the 2015 cohort (1st grade). In our tables and discussion, we refer to each of these points as 1st, 2nd, and 3rd waves.

In 2015 the Mexico City Office of Education denied test administrators access into control group public schools. Control group students were instead tested in their homes. CHM students were tested at the school. Varying testing conditions could by themselves explain some of the differences in test outcomes between treatment and control. To account for this potential source of bias, we run all of our analyses for two samples: (1) "All Observations," which includes the full sample of observations in all follow-up years, and (2) "Same Testing Conditions Sample," which includes only observations that were tested under the same conditions.

3.2.3. Learning outcome measures - instruments

Mexico has no national or state standardized tests in grades 1–3. For this reason, we use the EGRA (RTI International, 2009) to assess student skills in Spanish literacy and EGMA to assess student skills in mathematics (Mejia and Pflepsen, 2012). EGRA has been used to assess reading skills of children at grades 1–3 in more than 50 countries and 70 languages (Gove and Wetterberg, 2011). Prior studies showed it is a valid and reliable tool to measure reading achievement at early grade levels (RTI International, 2009).[7] EGRA includes eight subtests, including letter name knowledge, phonemic awareness, letter sound knowledge, listening comprehension, unfamiliar non-word reading, familiar word reading, passage reading and comprehension, and dictation. EGMA measures quantitative reasoning skills of students at grades 1–3. Prior studies showed that EGMA is a reliable measure of quantitative reasoning skills in primary schools (USAID, 2012). EGMA has seven subscales, including number identification, quantity discrimination, missing number, word problem, addition/subtraction problem, shape recognition, and pattern extension. Trained examiners administer these tests to individual students orally. Gains in EGRA/EGMA are measured by comparing results at the end of each corresponding grade per cohort. The average first grader is expected to score lower on the same test than the average third grader.[5]

We used RTI-approved Spanish translations of EGRA, which we adapted to Mexican-style Spanish and psychometrically validated in a pilot administration prior to the first year of data collection. Results from the validation exercise deemed both instruments to be reliable with a Cronbach's alpha of 0.88 for EGRA and 0.87 for EGMA (Barrios, 2014).

EGRA and EGMA tests were administered by an independent survey firm in Mexico City hired for this study. Neither the instruments nor the results of each wave of testing were shared with CHM. CHM had no access to any preliminary results before the last wave of data collection was completed, and thus had no influence on any of our results.

3.2.4. Parent survey data

We administered a survey to parents of students who participated in the 2014 lottery (Cohort 2 parents) to measure differences in opinions about their child's schooling and expectations as well as to collect data on out-of-pocket expenses incurred in their children's education. Out of 91 parents that consented to respond to the parent survey at baseline, 83 took the survey two years later, a 92% response rate. Response rate for treatment parents was 91%, and for control parents it was 94%. A comparison of baseline descriptive statistics for parents responding to the survey in treatment and control reveals a balanced sample with no significant differences (results not shown for brevity, but available upon request).

3.2.5. Cost data

We collected cost data about CHM from school staff in the Fall of 2016. Costs were only collected for the elementary school portion of CHM's operations. To gather cost data in control schools we used information provided by the Mexico City Office of Education, the Mexican Ministry of Education and other published sources. To fill in gaps and cross-reference the information provided by the Mexico City office of education we collected cost data from a sample of five control group schools. To estimate the average teacher cost in each of the control schools, we obtained data on the number of teachers and their seniority

[5] The EGRA design attends to the developmental nature of reading acquisition (Dubeck & Gove, 2015). Reading acquisition is a developmental process where higher-order skills (e.g., fluency and comprehension) build on lower-order skills (e.g., phonemic awareness, letter sound knowledge, and decoding). Five components are generally accepted as necessary to master the process of reading: phonological awareness, phonics (method of instruction that helps teach sound–symbol relationships), vocabulary, fluency, and comprehension (Armbruster, Lehr, & Osborn, 2003; Vaughn & Linan-Thompson, 2004) The EGRA subtasks (refer to Section 6) are aligned to these components (RTI International, 2016).

from the 2016–17 school census ("*Forma 911*"). Using these data, we identified each of the teachers and other staff working in control group schools by type of position. We used published salary schedules to estimate compensation for each teacher and staff position. We considered base salary and benefits, as well as any bonuses received by teachers participating in incentive programs such as "*Carrera Magisterial*.".

Most public schools in Mexico City received their buildings as donations from private individuals or foundations, and these donations go back many decades. The Mexico City Office of Education was unable to provide information that would allow us to estimate infrastructure costs except in the case of one school, which was being rented at market rates. We pro-rated the rent paid by this one school on the basis of enrollment and applied that rate to other control schools based on enrollment. Utility costs (water, electricity) were obtained using data provided by the Mexico City Office of Education for 83 public control group schools in our sample.

To estimate the cost of providing breakfast and lunch in schools where this service is available we used data from the school, teacher and student census of 2013 ("CEMABE 2013") as well as budget information for the National Free Breakfast Program. We also used information from CEMABE 2013 to identify control group schools that participate in the national breakfast program. Costs were pro-rated by school enrollment. To estimate the cost of providing lunch in full-day schools, we use data gathered from interviews with five control group schools to calculate per-student lunch costs.

We collected costs for educational textbooks and supplies. Textbook costs were obtained using the published cost of producing these textbooks by the National Commission on Free Textbooks ("*Conaliteg*"). Mexico has a national free textbook program, and all students in public elementary schools receive textbooks free of charge. To estimate school supplies costs, we use published prices from one large supplier in Mexico City. The Mexican Ministry of Education publishes an approved "school supply" list that schools in Mexico City adhere to. This list was used as a benchmark to estimate costs for all control group schools based on their enrollment. Both CHM and traditional public schools receive donated goods and services. In addition, parents pay out-of-pocket costs. We estimated these costs from the parent survey.

4. Methods

4.1. Academic outcomes and parental perceptions

We estimate analysis of covariance (ANCOVA) models that have the form:

$$Y_{ict} = \alpha_{cg} + \phi \quad Z_i + \beta \quad X_i^{'} + \epsilon_{ict} \tag{1}$$

where Y_{ict} is an outcome for applicant i in applicant cohort c (2013, 2014, 2015), measured in post-treatment year t (2014, 2015, 2016). As previously discussed we see Cohort 1 over 3 waves of data collection, Cohort 2 over two waves, and Cohort 1 over one wave. We include cohort-by-gender randomization strata fixed effects α_{cg}; Z_i is an indicator that takes the value of one if applicant i wins the admissions lottery and zero otherwise; X_i is the vector of baseline student background characteristics including applicant's age, RAVEN baseline cognitive score, mother's education, an indicator for single parent household, household income, and an indicator for whether the home is rented; a small number of observations (no more than 3%) had missing data on one or more the student baseline variables included in the model. To deal with this we impute missing values with the mean and include in the regressions a binary variable indicating whether values for that variable were imputed. To account for the fact that we have repeated post-treatment outcomes for the 2013 and 2014 applicant cohorts, we cluster the standard errors at the applicant level. Estimates of ϕ capture the regression-adjusted intent-to-treat (ITT) effect of being offered a first-grade slot at CHM (having won a spot in the lottery independent of whether the student took up the offer or not).

Because we observe different cohorts for a different number of years, we follow Angrist et al. (2010) to scale ITT impacts to annualized achievement gains. We do this by modeling the impact of CHM on outcomes as a function of years enrolled in the school:

$$Y_{ict} = \delta_{cg} + \pi s_{ict} + \rho X_i^{'} + \varepsilon_{ict} \tag{2}$$

where δ_{cg} are cohort-by-gender fixed effects; s_{ict} is the number of years applicant i from cohort c has spent in CHM at the time of the post-treatment measurement year t; X_i is the vector of student baseline background characteristics; ε_{ict} is a residual. We estimate equation (2) using instrumental variables in a two-stage least squares (2SLS) framework in which the randomly assigned lottery outcome Z_i is used to instrument for s_{ict}. This is done to prevent bias from the endogeneity of time spent at CHM (i.e. enrollment at CHM is not randomly generated, only the lottery assignment is randomized). Estimates of π capture the local average treatment effect (LATE) effect of each additional year in CHM. Coefficients (π) are interpreted as standard deviations (i.e. effect sizes).[6] Standard errors are clustered at the applicant level.

To test whether parents of lottery winners have different expectations and perceptions than parents of students who did not win the lottery, we modify Eq. (1) and (2) to include the relevant parental outcome as dependent variable.

4.2. Costs at the margin

To compare the costs of providing education at CHM compared to traditional public schools attended by control students, we use a version of the ingredients method, a simple yet rigorous procedure to guarantee inclusion of all relevant costs (Dhaliwal et al., 2011; Levin and McEwan, 2002, 2000). Using the data collected through primary and secondary sources we tally costs in three main categories: teacher payroll, total payroll (including administrative and support staff), and other costs, which include rent, public utilities, breakfast, lunch, computer and ICT equipment, and school materials (paper, etc.). These categories make up most of the operational costs of schools in our sample. One potentially important cost that we do not include is parental time, which is required at CHM and frequently donated at public schools across the country (Santibañez et al., 2014). We do not have data to estimate the monetary value of parental time. Given that over 90% of the cost of public schools in Mexico covers teaching and non-teaching staff salaries we feel confident our estimation is close to the real cost of providing schooling to students in our sample. Itemized data on out-of-pocket expenses was averaged separately for CHM and control group parents.

5. Internal validity: balance and attrition

To determine the internal validity of our experiment, we follow the U.S. Department of Education's Institute for Education Sciences What Works Clearinghouse (WWC) guidelines. The WWC was created to guide researchers and educators in making evidence-based decisions. To do this, it conducts systematic reviews of studies to determine whether their findings are valid and can be trusted as a source of scientific evidence. The WWC publishes the procedures and benchmarks it uses to determine that a study meets scientific standards. In the case of experimental studies, WWC determines that a study meets evidence standards

[6] EGRA/EGMA yield results in multiple domains (nine subscales for EGRA and seven for EGMA). To avoid presenting 16 different equation results, we pool all subscale effects into a single effect size using a fixed-effects model (Coope et al., 2009; Bettinger et al., 2010; Katz et al., 2001). To do this, first we standardize all domain score results relative to the control group mean and standard deviation (SD) in the first follow-up. Then, we calculate a precision-weighted average effect size. This is obtained using the inverse of the estimate's variance and calculating its t-statistic (Cooper et al., 2009).

if it meets two conditions: treatment and control group membership is determined through a random process and the combination of overall and differential sample attrition is low (WWC, 2020). As noted earlier, treatment and control group membership in the study were determined through publicly-conducted lotteries. We can be certain that our lottery procedure worked since differences between treatment and control in baseline demographic and cognitive measures are not significant for any of the variables used in the analysis both for the whole sample, as well as for the sample of students tested under the same conditions (see Table A2 and A3 in the Online Appendix).

Overall sample attrition in our study was low. Initially, 242 students participated in the lottery. Testing students throughout the duration of the study would have yielded a total of 520 potential observations (See Table 2, Panel A). While no student missed all data collection events, 18 observations do not have outcome data for a total analytic sample size of 502 observations. This indicates an overall attrition rate of 3.46% and differential attrition of 3.74% (Table 2, Panel B). These low overall and differential attrition rates are well within the benchmarks issued by the Institute of Education Sciences' WWC for evidence in randomized experiments: 6% differential attrition when the overall attrition rate is 4% (WWC, 2020).

To further examine differential attrition, we conduct two tests based on Duflo, Glennerster and Kremer (2007). Test 1 examines whether the fraction of applicants with missing outcome data statistically differs between lottery winners and losers. Test 2 examines whether the factors that predict attrition are the same between winners and losers.[7] Finding statistically significant results on any of these two tests raises concerns about differential attrition. Results of these analyses are found on Table 3. Test 1 finds no differential attrition at conventional levels (95%) by follow-up point (wave), but does find statistically significant differential attrition with the pooled sample. Test 2 fails to find significant differential attrition as jointly predicted by baseline characteristics.

Table 2
Overall and differential attrition.

Panel A: Baseline vs. Analytic Sample						
	T	C	Total			
Baseline Sample Size (Students)	100	142	242			
Potential Analytic Sample Size (without attrition)*	187	333	520			
Analytic Sample (including attrition)*	185	317	502			
*Over three waves of the study						
Panel B: Attrition Counts and Proportions, Overall and by Wave						
	Overall Attrition	%	T	%	C	%
Overall	18	3.46%	2	1.07%	16	4.80%
Differential attrition						3.74%
Wave 1	3	1.26%	0	0.00%	3	2.16%
Wave 2	9	5.14%	1	1.67%	8	6.96%
Wave 3	6	6.82%	1	4.00%	5	7.94%

[7] Test 1 is done by regressing A_i, an indicator for whether outcome data is missing for applicant i, on lottery status, Z_i lottery stratification indicators ϑ_{cg} (cohort-by-gender), and baseline covariates for each applicant cohort. We estimate regression equation (3) separately at each follow-up point since attrition patterns can vary over time. Test 2 is similar to test 1, but includes interactions between Z_{ic} and the vector of baseline covariates.

Table 3
Differential sample attrition tests.

	Wave 1	Wave 2	Wave 3	Pooled
Test 1: Differential Attrition Test/1				
Lottery Assignment	-0.028	-0.042	-0.042	-0.035 * *
	(0.016)	(0.025)	(0.049)	(0.014)
N	242	184	94	520
Test 2: Predictors of Differential Attrition/2				
Lottery Assignment	0.368	-0.130	-0.265	-0.026
	(0.078)	(0.366)	(0.457)	(0.140)
F-test for interactions (p-value)	0.878	0.599	0.439	0.262
N	242	184	94	520

* * $p < 0.05$, *$p < 0.10$
/1 The first test examines whether the fraction of applicants with missing outcome data statistically differs between lottery winners and losers.
/2 The second test examines whether the factors that predict attrition are the same between lottery winners and lottery losers. It calculates an F-statistic to test the null joint hypothesis that all the parameters in the model are jointly zero.
Note: All models are run using OLS regression and include lottery strata (cohort-by-gender dummies) and student-level demographic and other covariates. Results don't differ if these are run as linear probability models using logistic regression. Errors are clustered at the student level.

Because we find mixed evidence of moderate differential attrition, we turn once again to the WWC for further guidance. The WWC suggests that in the presence of differential attrition researchers should consider the type of sample and the likely relationship between attrition and the study outcome. When the reason students do not have outcome data can be traced to children's residential or school mobility or regular absence from school and unrelated to the actual treatment, sample attrition is considered less of a threat to the study's internal validity. This is the case in this study. During Wave 2 (where attrition was highest for the control group) we were unable to find several children because they were not present at home on the day of the test. In other waves, some children had moved out of state. Taken together, these findings make us confident that our study is internally valid as defined by the WWC Standards (WWC, 2020).

6. Results

6.1. Descriptive analysis

Table 4 shows descriptive statistics on key demographic and socioeconomic variables for CHM applicant families (by lottery offer/treatment status) and the average household with students in public school in Mexico City. CHM applicant families are more disadvantaged socioeconomically than the average household in Mexico City with children in public schools. CHM applicant families earn about half what those in the average Mexico City household sample earn annually—about $US2,400-$2700 compared to $4700 for the Mexico City sample (exchange rate 20 pesos/dollar). Overall, CHM applicant mothers have slightly less schooling than the household sample, but in both cases the average level of schooling completed is lower secondary school (9th grade). It should be noted that Mexico City has one of the highest levels of average educational attainment in the country. Fewer students live in homes with both parents present in the CHM applicant sample than in the Mexico City household sample, about 51% and 75% respectively. Lastly, CHM applicant families are more likely than the average Mexico City family to rent their home, not have a kitchen, not be hooked to a public utility source for electricity, and not to have a landline.

6.2. Impacts on early-grade literacy and numeracy

Table 5 shows intent-to-treat (ITT) effects of winning the admissions

Table 4
Descriptive statistics and balance.

Variable	CHM-Offered Admission (Treatment) (1)	CHM-Not Offered Admission (Control) (2)	Difference (C-T) (4)	Standardized Difference[/3] (4)	Students in Mexico City Public schools (5)
Annual income $USD[/2]	2399	2628	19.088	0.349	4687
	-160.46	-120.54	(20.333)		-251.4
Mother's education (Secondary=4)	4.24	4.144	-0.096	0.311	4.35
	-0.068	-0.063	(0.095)		-0.073
Both parents are present in the home	0.52	0.507	-0.013	0.845	0.75
	-0.05	-0.042	(0.065)		-0.031
Home is rented	0.44	0.402	-0.038	0.558	0.24
	-0.05	-0.041	(0.064)		-0.03
Home has a kitchen	0.59	0.67	0.080	0.199	0.95
	-0.049	-0.039	(0.062)		-0.016
Observations	*139*	*100*			*270,000*
P-value of differences in column (4) are noted by: * * p < 0.05					

Notes: Source for Mexico City sample is ENIGH-2014 (tradicional). HH with children in grades 1–3 in public schools. Sample represents over 270,000 HHs in Mexico City. Source for CHM treatment and control is lottery sample. F-value tests the hypothesis that the coefficients on key demographic variables measured at baseline (age, Raven, WISC) on winning the lottery (treatment status) are all jointly zero. F-value from joint test: 0.71. p-value from F-test 0.54.
/1 Student Age for CHM is collected between March-June in the school year previous to entering 1st grade.
/2 Average monthly household income from wages and other fixed sources.
/3 Standardized with respect to the control group standard deviation

Table 5
ITT estimates of the impacts of CHM lottery offer on early-grade outcomes over three years.

	(1)	(3)	(4)	(2)
Panel A. Early Grade Literacy (EGRA)				
Won admissions lottery	0.18 * *	0.17 * *	0.22 * *	0.22 * *
	(0.032)	(0.030)	(0.036)	(0.034)
Panel B. Early Grade Numeracy (EGMA)				
Won admissions lottery	0.09 * *	0.07 *	0.17 * *	0.16 * *
	(0.038)	(0.037)	(0.042)	(0.041)
Sample[/1]	All Observations	All Observations	Comparable testing conditions	Comparable testing conditions
Demographic and cognitive baseline controls[/2]	No	Yes	Yes	No
Observations (N)	502	502	328	328

* * p < 0.05, *p < 0.10
Note: All regressions include lottery strata fixed effects (gender-by-cohort). Panel A are intent-to-treat effects with lottery offer as the key explanatory variable. Standard errors clustered at the student level in parentheses. Controls include: age, gender, cohort, RAVEN test scores at baseline, mother's education, single parent household, monthly household income, home is rented. Missing values imputed, and missing dummies included for imputed cases.

lottery. Results show that CHM students who won a seat in the school's first grade cohort via lottery scored 0.18 SD higher in early grade literacy, and 0.09 SD higher in early grade numeracy than students who did not win the lottery. When students were tested under comparable conditions, the coefficients are 0.22 SD in literacy and 0.17 SD in numeracy. All of these effects are statistically significant at conventional levels. The magnitude of the effects remains virtually unchanged when student controls are included.

To scale the ITT estimates into annualized gains, we instrument the number of years that a student spent at CHM during the sample period with lottery offers using 2SLS. The first-stage results, shown in Table 6, indicate that winning an offer of admission results in about two additional years spent in CHM. The 2SLS results, shown in Panels B & C, indicate that students gain an average of 0.10 SD in literacy for each year spent in CHM and 0.05 SD in numeracy. When the sample includes only students tested under comparable conditions, each additional year in CHM raises literacy scores by 0.13 SD and numeracy scores by 0.10 SD. All reported results are statistically significant at the 95% confidence level. To put these effects into context, we compare them with the average growth of the control group: 0.45 SD per year in literacy and 0.71 SD per year in numeracy. Using the lower bound estimates (0.10 SD) means that after two years the average student at CHM would be about 20% ahead of the control group —slightly over 2 months. Gains in numeracy are very small.

6.3. *Effects on parental perceptions and expectations*

Parents of students who win the lottery of admission at CHM exhibit more satisfaction and higher expectations for their children than control-group parents. This is consistent with private and charter school literature which finds private school students are more satisfied with their school (Rhinesmith, 2017; Barrows et al., 2019). CHM parents think their school is very demanding academically and stricter—to a greater degree than control parents (see Table 7). On a scale of 1–10, CHM parents rate their school a 9.8 vs. 8.2 for control group parents. CHM parents have higher expectations for their child's education than parents in the control group. On a scale of 1–10, control group parents set the probability of their child completing high school at 9.5 vs 9.2 in the case of control group parents, but this difference is not statistically significant. In contrast, CHM parents have significantly higher expectations that their children will complete college: 9.5 (on a scale of 1–10) relative to 8.6 in the control group– almost a whole-point difference. Results are consistent with and without baseline covariates.

6.4. *Robustness tests*

When randomizing subjects into treatment and control groups, findings can be sensitive to just a few outliers or clusters of observations. Young (2018) and Athey and Imbens (2017) show that model-based regression designs may produce cluster and robust variance estimates that are unstable for a given regression design generating incorrect tests

Table 6
Annualized early-grade outcome gains from attending CHM (IV Results).

	(1)	(2)	(3)	(4)
Panel A. First Stage				
Won admissions lottery	1.85 * *		1.81 * *	
	(0.131)		(0.136)	
Panel B. Early Grade Literacy (EGRA)				
Years in CHM (instrumented)	0.10 * *	0.09 * *	0.13 * *	0.12 * *
	(0.017)	(0.016)	(0.019)	(0.018)
Panel C. Early Grade Numeracy (EGMA)				
Years in CHM (instrumented)	0.05 * *	0.04 * *	0.10 * *	0.09 * *
	(0.020)	(0.019)	(0.022)	(0.021)
Sample/1	All Observations	All Observations	Comparable testing conditions	Comparable testing conditions
Demographic and cognitive baseline controls/2	No	Yes	No	Yes
Observations (N)	502	502	328	328

* * $p < 0.05$, *$p < 0.10$
Note: All regressions include lottery strata fixed effects (gender-by-cohort). Panel A are intent-to-treat effects with lottery offer as the key explanatory variable. Standard errors clustered at the student level in parentheses. Controls include: age, gender, cohort, RAVEN test scores at baseline, mother's education, single parent household, monthly household income, home is rented. Missing values imputed, and missing dummies included for imputed cases.

Table 7
Effects of CHM lottery results on parental perceptions and expectations.

	Control Mean (1)	Won Admissions Lottery (ITT) (2)	Enrolled in CHM (IV) (3)
My school is very demanding academically (1–4)	2.67	0.89 * *	0.49 * *
		(0.163)	(0.077)
My school is very strict (discipline) (1–4)	2.85	0.82 * *	0.45 * *
		(0.162)	(0.076)
I think my school is (1 =Bad, 4 =Excellent)	2.88	1.03 * *	0.57 * *
		(0.132)	(0.058)
On a scale of 1–10, my school is a….	8.21	1.61 * *	0.89 * *
		(0.241)	(0.108)
On a scale of 1–10, the probability my child will complete high school is.	9.23	0.23	0.13
		(0.196)	(0.098)
On a scale of 1–10, the probability my child will complete college is.	8.57	0.96 * *	0.53 * *
		(0.248)	(0.132)
Sample		Cohort 2 Parents Only	
Demographic and cognitive baseline controls/1		Yes	Yes
Observations (N)		83	83

* * $p < 0.05$, *$p < 0.10$
/1 Controls include: age, gender, cohort, RAVEN test scores at baseline, mother's education, single parent household, monthly household income, home is rented. Missing values imputed, and missing dummies included for imputed cases.
Note: All regressions include lottery strata fixed effects (gender-by-cohort). These are intent-to-treat effects with lottery offer as the key explanatory variable, and questions about perceptions as the dependent variable in each separate regression. Standard errors clustered at the student level in parentheses.

of significance. To address this potential issue, we estimate randomization p-values and omnibus randomization inference tests for all of our analyses (academic outcomes, parental outcomes).

Table 8 shows results from randomization inference omnibus tests performed on the joint significance of test score outcomes. The literacy results are robust to randomization inference (mean randomization p-value of 0.001 in the omnibus test). The numeracy results are not, suggesting that some observations (i.e. a few outliers or clusters) are causing volatility in the standard errors and so the significance tests of these estimates are inaccurate (not robust). Parental outcome results are also robust to this test.

One concern with our focus on a single school, is that the population of the school may not resemble the "average" family in Mexico City. As

Table 8
Randomization inference p-values and outcome joint-significance omnibus tests.

Omnibus Tests	Pooled Min p-value	Max p-value	Randomized p-value	N
Early Grade Literacy				
Sample 1 - all observations	0.001	0.002	0.002	502
Sample 2 - comparable testing conditions	0.000	0.001	0.001	325
Early Grade Numeracy				
Sample 1 - all observations	0.239	0.240	0.239	502
Sample 2 - comparable testing conditions	0.234	0.235	0.234	325
Parent percpections of how demanding the school is	0.000	0.001	0.001	83
Parents opinion of the school (scale 1–10)	0.000	0.001	0.001	83
Parent expect student to go to college	0.001	0.002	0.002	83

Notes: Randomization inference tests are conducted using *randcmd* in STATA (Young, 2017). 1000 permutations are conducted for each test.

shown in Table 4, families applying to the CHM lottery look different from the average family in Mexico City. This could result in "participation bias" (Andrews and Oster, 2018). To address this we use the method of "entropy weights" as proposed by Hainmueller (2012) and Hainmueller and Xu (2013) to re-weight the sample to resemble the average Mexico City family.[8] As shown in Table A4 in the Online Appendix, the positive impact of CHM on literacy effects holds (with a larger magnitude in the reweighted exercise), and the numeracy results remain non-significant.

6.5. Costs and cost-effectiveness

Table 9 shows estimates of annual CHM relative to counterfactual elementary public schools attended by students who did not win the CHM lottery. The annual (10-month) total cost of educating a student at

[8] The method resembles propensity score matching. To the extent that most of the potential participation bias is explained by observable characteristics, results will be externally valid (Andrews and Oster, 2018). To estimate the weights, we use Mexican household data from the *Encuesta Nacional de Ingreso y Gasto de los Hogares* 2014 (ENIGH-MCS). We select households in Mexico City with children attending public schools in 1st-3rd grade and use variables that overlap with socioeconomic data from our baseline. The average weights closely replicate the first and second moments of the Mexico City household distribution (results not shown, but available upon request).

Table 9
Ingredients-method cost analysis ($ per pupil/year).

	CHM[1]	All Public	Half-Day	Full-Day
Operation Costs				
Teacher Payroll[2]	$317	$433	$365	$517
All Payroll (incl. staff)[2]	$1119	$690	$559	$847
Other Costs[3]	$1284	$537	$267	$448
Total Per-Student Operation Cost (All Payroll + Other Costs)	*$2403*	*$1227*	*$826*	*$1295*
Parent out-of-pocket Costs[4]				
Lump-sum costs (one time)				
Matriculation Fees	$0	$5	$7	$3
Uniforms	$3	$41	$45	$38
Textbooks, materials, supplies	$1	$24	$23	$24
Other school expenses[5]	$10	$93	$103	$84
Annualized Recurring Costs[6]				
Meals	$7	$157	$133	$181
Voluntary Fees or contributions (incl. tuition if required)	$131	$111	$127	$94
Transportation	$237	$166	$114	$218
Textbooks, materials, supplies	$23	$72	$92	$52
Total Per-Student Average Out-of-Pocket Expenses (annual)	*$412*	*$668*	*$644*	*$693*
TOTAL (Operations + Parents Out-of-Pocket)	*$2814*	*$1895*	*$1470*	*$1988*

Note: Exchange rate is 18 pesos per dollar. The control group includes 83 public schools with available data. Elementary enrollment is 337 for CHM. Average enrollment is 361 for all public schools in control group, 377 for half-day schools, and 357 for full-day schools.
1) Source for CHM costs are staff interviews and documentation provided by institution.
2) Payroll estimates for public schools are based on the school census ("Forma 911") for the 2016–2017 school year, and the salary schedule published by AFSEDF for the same school year. Salaries include incentives (Carrera Magisterial), benefits (christmas bonus "aguinaldo", vacation premium and other benefits.) Amounts for benefit calculations were obtained from INAI.
3) Other costs include rent, public utilities, breakfast, lunch, computer and ICT equipment, and school materials (paper, etc.). They do not include costs for regular maintenance, textbooks, school uniforms and any other expenses. Assumptions for these other costs: Rent for public school is estimated based on AFSEDF calculations for the one school that actually pays rent, pro-rated by student enrollment in each building. All other schools are operating in buildings donated by private individuals, in buildings being borrowed from the municipality or in other situations for which rent cannot be estimated. Public utility costs are only calculated for water and electricity, since no other information was provided by AFSEDF. For nutrition, information comes from CEMABE 2013–14 (national breakfast program) and full-lunch program (Escuelas de Tiempo Completo only).
4) Responses are for Cohort 2014 parents. Survey was administered in June of 2015, and again in June 2016. Responses are the average of both responses. Parents were asked to refer to the study sample child only. Parent responses were coded corresponding to their actual child's enrollment (not lottery-status). This was done in order to get accurate estimates of average costs in CHM vs. public schools.
5) Other school expenses include: school festivales, field trips, carnival, pictures, extended day, and music lesson fees.
6) Monthly expenses such as transportation, lunch, school materials, tuition and fees, were converted to yearly costs using a 10-month academic school year.

CHM (including operating costs + parent out-of-pocket costs) in US dollars is $2814 per year. The per-student annual cost in comparison public schools is US$1895. This is US$1470 for half-day traditional public schools, and US$1988 for full-time public schools (*jornada ampliada* and *tiempo completo*).[9]

[9] The operating costs are in line with published sources. The national evaluation institute in Mexico, INEE, calculates an average annual (10 month) per-student cost of US$900 for elementary public schools in the country (INEE, 2017). Costs in Mexico City are likely to be higher than the national average owing to cost of living.

Teacher salaries do not drive the higher per-pupil cost of CHM. In fact, the average teacher payroll at CHM is US$317 per student, about 27% lower than the teacher bill for control group public schools (US $433 yearly per student for all schools, US$365 for half-day schools, and US$517 for full-day schools). As previously discussed, CHM is able to do this by drawing from a pool of private teacher-college graduates or retired public-school teachers. CHM spends more per student in non-teaching staff (counselors, psychologists, administrators, etc.). The total payroll cost including teaching + non-teaching staff at CHM is US $1119 per student, while at all public schools it is US$690 (US$559 for half-day schools, US$847 for full-day schools). When "other costs" are included (i.e. rent, utilities, nutrition, uniforms, extracurricular activities, enrichment, extended day, etc.) are added to staffing costs, CHM has significantly higher total operation costs: US$2403 vs. US$1227 for all public control schools ($826 on average for half-day schools, US $1295 for full-day schools).

At CHM, 53% of per-student operating costs are in the "other" cost category which includes rent, public utilities, breakfast, lunch, computer and ICT equipment, and school materials; 33% are for non-teaching staff, and 13% for teaching salaries. In control public schools, 44% of per-student operating costs are in the "other" category, 21% are for non-teaching staff, and 35% for teacher salaries. This suggests CHM invests heavily on student services (beyond instruction). In addition, because it is only one school, CHM has few economies of scale that would result in efficiency gains in some expense categories.

Because CHM spends considerably higher than public schools per student, parents spend less money out of pocket for school-related expenditures including matriculation fees, school uniforms (which are required for public schools and CHM), textbooks, school supplies, and nutrition. The only category in which CHM parents spend significantly more money than public school parents is in transportation to and from school.

The achievement gains of the CHM schooling model come at an increased cost relative to control public schools of about US$1000 per pupil/year. Recall that the robust per-year effect of enrollment in CHM had an impact of 0.10 SD in literacy. Relative to other educational interventions in low-income countries, this ratio suggests low cost-effectiveness (Evans & Ghosh, 2008) such as class size reduction or remedial education (US$10 and US$300 per 0.10 SD improvement in test scores respectively). It should be noted these studies were conducted in the late 1990 s and early 00 s, so these costs may be outdated relative to this case-study.

7. Concluding remarks

This paper uses a lottery-based admissions process for a philanthropic, comprehensive private school model in Mexico City to estimate CHM's effects on early-grade literacy and numeracy outcomes. It is one of a handful of rigorous studies to investigate whether a philanthropic private school model can be a cost-effective alternative to educate low-income children in Mexico and other developing countries.

We find robust evidence to suggest that children living in urban poverty in Mexico City perform better on early grade literacy tests when they attend CHM relative to traditional public schools. Numeracy effects are also positive, but not robust. In addition, parents of children who win the CHM admissions lottery express higher levels of satisfaction with the school and are more likely to expect that their child will complete college than parents of children who did not win a spot through the lottery.

On a per-student basis, CHM is not a "low-cost" model such as the schools that are frequently the focus of research. The services offered by CHM cost about US $1000 more than the services offered by control public schools in Mexico City. Because it is a single school, CHM cannot take advantage of economies of scale available to public schools. In addition, CHM invests heavily in "wrap-around" services that students and their families need to be successful and remain in school—services not usually available in public schools. While early-grade learning outcomes are foundational and predictive of future academic achievement, the long-term positive effects of many of these services could be even stronger. For example, higher early-grade outcomes could translate into higher school attainment. At CHM 98% of CHM students finish high school, compared to only 30–40% of upper secondary students in the bottom two income quartiles in Mexico—the comparable peer population (Josephson et al., 2018). Higher school attainment has consistently been found to predict better long-term employment and earnings (Kugler and Rojas, 2018; Behrman et al., 2011; Chetty et al., 2011).

Philanthropically-funded comprehensive private schools like CHM could significantly improve learning among Mexico's lowest-income children. There is clear demand in the region for private schooling (Elacqua et al., 2018). And while no hard data exist on supply, i.e. the numbers of students enrolled in non-profit, philanthropic private schools, there is clearly a group of people invested in supporting and growing these institutions. For this study we surveyed more than 60 schools enrolling over 30,000 students in Mexico City (see Table A1 in the online appendix). Latin America has large networks such as *Fe y Alegria* (close to 1 million students in over 15 countries in Latin America (Osorio & Woodon, 2011), privately-run *Escuela Nueva* models in Guatemala and other countries (Kline, 2002), and concession schools in Colombia just to name a few.

This case study has clearly shown that it is possible to significantly raise educational achievement for very low-income children—an important and unresolved issue in the region. Given the deep and persistent inequalities in Mexico and many other developing countries, more research is needed on the long-term benefits of these kinds of schools, their prevalence in the region, as well as their cost- and other conditions that could affect scale-up efforts. A better understanding of these issues could lead to improved academic and other outcomes for the region's most vulnerable children.

CRediT authorship contribution statement

Lucrecia Santibañez: Conceptualization, Methodology, Investigation, Writing – original draft, Writing – review & editing, Supervision, Funding acquisition. **Juan E. Saavedra**: Conceptualization, Methodology, Investigation, Writing – review & editing, Funding acquisition. **Raja B. Kattan**: Writing – original draft, Project administration, Funding acquisition, Supervision. **Harry A. Patrinos**: Conceptualization, Writing – original draft, Writing – review & editing, Funding acquisition.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ijedudev.2021.102494.

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